Cropping treatment effects and growth legacy

Analysis notes for paper drafting | Updated 22 October 2026

This note records the PAIC comparison among continuously cropped, alternately cropped and continuously fallow alleys, during cultivation and after cessation. Continuous cropping reduced PAIC during active cultivation. After cessation, a temporary growth advantage over former continuous fallow is suggestive of release from crop competition, with the strongest interval-specific evidence in growth years 2008 and 2009.

Experimental design and response

Within each of three blocks, six alleys were allocated in adjacent pairs to the three cropping treatments. A measurement row lay between each paired set of alleys. Continuous fallow had both alleys fallow; continuous cropping had both cropped; alternate cropping had one cropped and one fallow, reversed each year. Alternate cropping is not a second wholly fallow treatment.

All available measured trees are retained. There are nine treatment measurement rows with five trees of each of four hybrids per row, giving 180 tree IDs. The design describes guard trees at row ends. Each row by hybrid by interval mean is equally weighted in the model; the five neighbouring trees are subsamples, not independent treatment replicates.

Growth intervals and cessation

Cropping stopped after the 2004 harvest. The primary active model includes observed intervals through the 2002 visit. The primary post-cropping model begins with the 2005 to 2006 interval, representing growth year 2005. The 2002 to 2005 boundary interval includes growth years 2002 to 2004 under the adopted timing convention. It is excluded from both primary fits and included with active cropping in a sensitivity analysis.

The initial visit has no preceding increment and is excluded. The low PAIC at the 2001 visit is a three-year average across growth years 1998 to 2000. Missing visits limit temporal resolution; they do not themselves cause low annualised growth.

Statistical model

Separate Gaussian mixed models include fixed block and the full treatment by hybrid by interval interaction. Random intercepts represent treatment row, row by hybrid and row by interval. Block remains a design adjustment irrespective of significance. Treatment comparisons average equally across hybrids and observed intervals; this differs from duration-weighting the intervals.

Overall treatment contrasts use Bonferroni adjustment for three pairs. Interval-specific inference uses Holm adjustment across all three pairs and all intervals within each phase: 21 active, 54 post-cropping and 24 boundary-sensitivity tests. Confidence intervals in the marked figures are pointwise; markers use the phase-wide Holm tests. The three fits are not pooled into a single multiplicity family because the primary phases address separate questions and the third fit is a sensitivity analysis.

Model specification and block adjustment

The following pseudo-R specification mirrors analysis/03_treatment_active.R. Fit it separately to the active and post-cropping subsets, and repeat with the boundary interval included in active cropping for sensitivity. PAIC is the equally weighted mean across available measured trees within each row, hybrid and interval. Block, treatment, hybrid and interval are factors.

```
# IDs for the row by hybrid by interval means
dat$row_id <- interaction(dat$block, dat$tree_row)
dat$row_hybrid <- interaction(dat$row_id, dat$hybrid_code)
dat$row_interval <- interaction(dat$row_id, dat$interval)

fit <- lmerTest::lmer(
  PAIC ~ block + treatment_code * hybrid_code * interval +
    (1 | row_id) + (1 | row_hybrid) + (1 | row_interval),
  data = dat, REML = TRUE,
  contrasts = list(block = "contr.sum",
    treatment_code = "contr.sum", hybrid_code = "contr.sum",
    interval = "contr.sum")
)
anova(fit, type = 3, ddf = "Satterthwaite")
emmeans(fit, ~ treatment_code | interval, weights = "equal",
        lmer.df = "satterthwaite")
# Holm adjustment spans all three pairs and all intervals
# within a phase; see treatment_presentation.R.
```

The asterisks in the formula expand all main effects and interactions. The terms beginning with 1 | are random intercepts: row for persistent differences among treatment units, row by hybrid for persistent local hybrid differences, and row by interval for fluctuations shared by hybrids in a row. Block is a fixed adjustment; it is distinct from these random terms.

Tests of the block effect

Model	F statistic	Degrees of freedom	p value
Active cropping	0.798	2 and 4.00	0.511
After cropping ceased	0.481	2 and 4.00	0.650
Active including boundary	0.458	2 and 4.00	0.662

These are Type III F-tests with Satterthwaite denominator degrees of freedom from the fitted models. They provide no clear evidence of mean block differences, but do not demonstrate that blocks are identical.

We retain block because it is part of the allocation design. Its purpose is to adjust treatment comparisons for between-block differences, not to identify a substantive effect for the paper. Removing it only when its test is nonsignificant makes the inferential model depend on a preliminary test. The three blocks remain the treatment replicates; retaining block does not turn individual trees into extra independent replicates.

Methodological reference: Frey, Hartung, Ogutu and Piepho (2024), Analyze as randomized: Why dropping block effects in designed experiments is a bad idea. Agronomy Journal 116, 1371-1381. https://doi.org/10.1002/agj2.21570

Main results

Phase	Treatment difference	PAIC difference	Adjusted p
Active	Continuous cropping - Continuous fallow	-0.473	0.0068
Active	Alternate cropping - Continuous fallow	-0.190	0.1475
Active	Continuous cropping - Alternate cropping	-0.283	0.0428
Post	Continuous cropping - Continuous fallow	0.310	0.1026
Post	Alternate cropping - Continuous fallow	0.127	0.7920
Post	Continuous cropping - Alternate cropping	0.183	0.4084

PAIC differences are Mg C per hectare per year. The active treatment omnibus p-value is 0.0057. The post-cropping treatment omnibus p-value is 0.0806; the average post-cropping pairwise contrasts are uncertain and do not exclude shorter-lived differences.

Including the boundary interval gives continuous cropping minus continuous fallow of -0.445, adjusted p = 0.0075. The corresponding continuous cropping minus alternate cropping contrast is -0.260, adjusted p = 0.0506. The continuous-cropping versus continuous-fallow conclusion is consistent; the other active average contrast is sensitive near the conventional threshold.

Differences among hybrids

Hybrid main effects are significant in both primary models, with p below 0.001. Treatment by hybrid interactions are nonsignificant: active p = 0.613 and post p = 0.570. The three-way interactions with interval are also nonsignificant. We therefore display treatment comparisons averaged across hybrids, without claiming that hybrids have equal PAIC or identical treatment responses.

Post cropping intervals surviving adjustment

Growth year	Comparison	Difference	Holm p
2008	Continuous cropping - Continuous fallow	0.809	0.0024
2009	Continuous cropping - Continuous fallow	0.642	0.0380

These correspond to the 2009 and 2010 visits. The broader positive pattern over growth years 2007 to 2010 is descriptive, not significant in every year after phase-wide correction. The growth-year-2013 difference does not survive correction; no interval has been removed as a presumed fluke.

Files supporting this note

analysis/03_treatment_active.R

scripts/treatment_presentation.R

analysis/03b_refresh_treatment_figures.R

results/tables/treatment_PAIC_*_overall_contrasts.csv

results/tables/treatment_PAIC_*_interval_inference.csv

Growth during active cropping

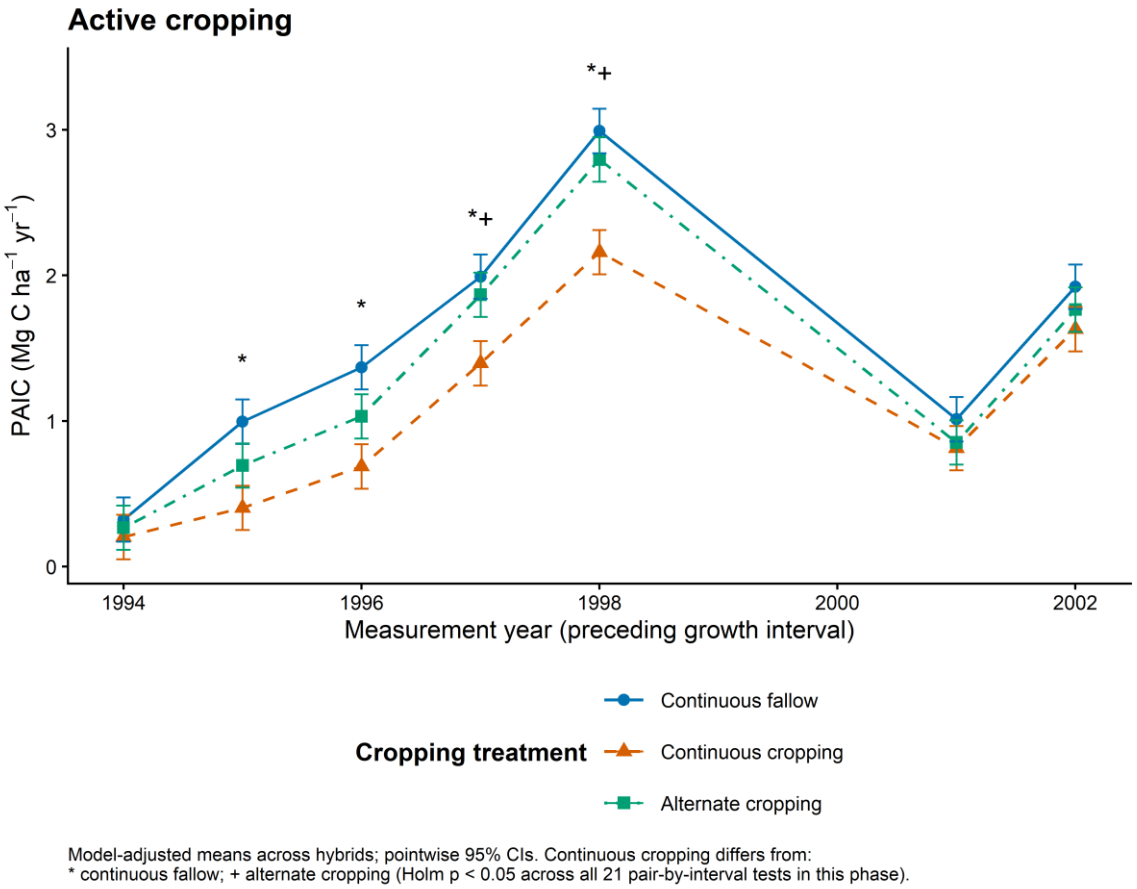


Figure 1 Model-adjusted PAIC during active cropping, averaged equally across hybrids. Asterisks indicate continuous cropping versus continuous fallow differences; plus signs indicate continuous cropping versus alternate cropping differences, after Holm correction across all 21 pair-by-interval tests. Error bars are pointwise 95 percent confidence intervals.

The x-axis gives measurement years, not single annual growth years. In particular, the 2001 point averages growth years 1998 to 2000. The three-year measurement gap prevents us from identifying the annual trajectory within that interval; it does not itself explain the low annualised mean. Nonsignificant contrasts at the late visits are not evidence of equal growth.

Discussion of the post cropping legacy

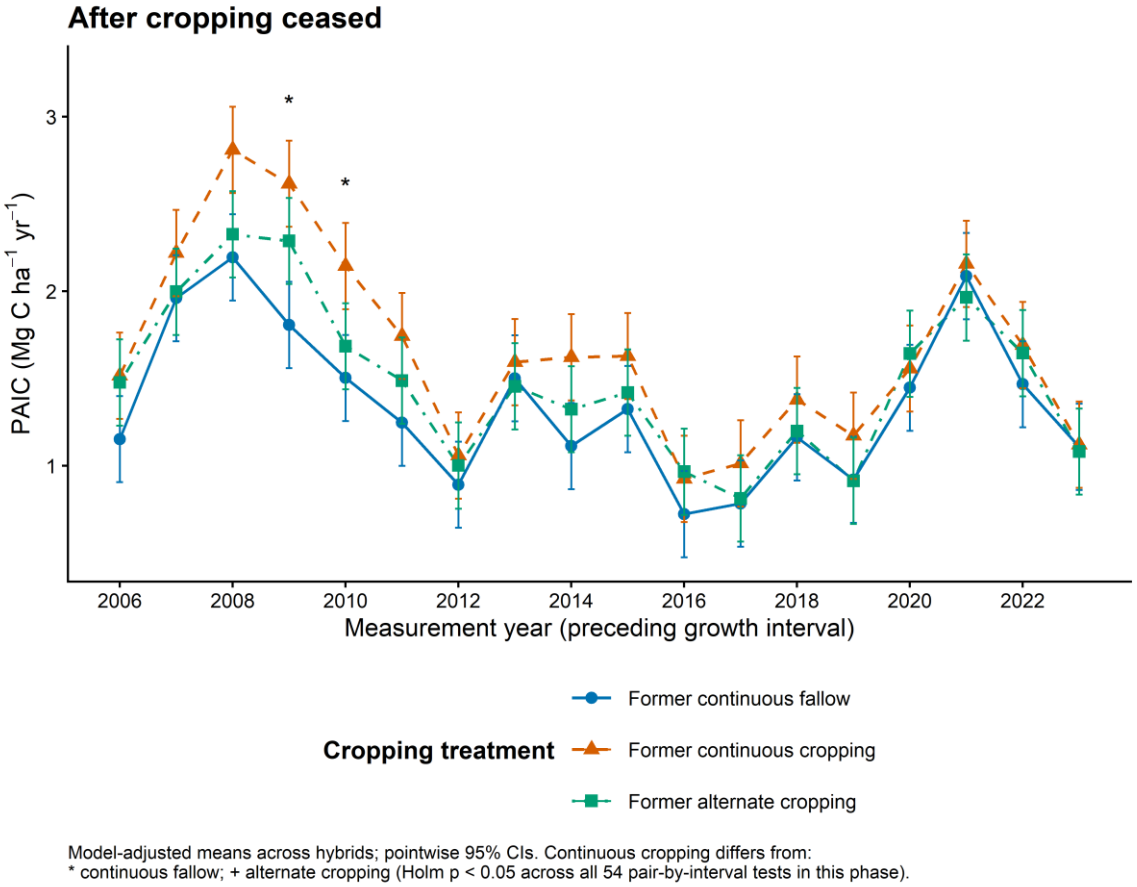


Figure 2 Model-adjusted PAIC averaged across hybrids after cessation. Asterisks mark continuous cropping versus continuous fallow differences after phase-wide Holm correction. Error bars are pointwise 95 percent confidence intervals.

Suggested drafting language: Following cessation of cropping, trees formerly under continuous cropping showed a temporary PAIC advantage over trees formerly under continuous fallow, with the strongest statistical evidence in growth years 2008 and 2009. This pattern is consistent with release from crop competition, but does not establish that mechanism.

Post-cropping contrasts include consequences of earlier differences in size and condition. Weather, stand development and other temporal influences may contribute to the trajectories. A nonsignificant contrast does not date disappearance of a legacy or demonstrate equivalence between treatments.

Active cropping convergence and remaining decisions

During the 1998 to 2001 interval, PAIC fell in all treatments and the treatment differences narrowed. Continuous fallow minus continuous cropping declined from about 0.83 at the 1998 visit to 0.20 at the 2001 visit, then rose to 0.29 at the 2002 visit. The weather explanation remains untested with interval-aligned predictors.

All current fits converged without singularity. Residual plots show some unequal spread and tail departures; variance assumptions and remaining serial dependence require review before final inferential wording. Three blocks limit independent treatment replication. A biological equivalence threshold would be needed to estimate when a legacy becomes negligible.